

## Threats in IoT enabled architecture

As per the Singapore Computer Society (SCS) Cyber Literacy Report 2020 and Blackberry Threat Report 2022, IoT attacks remain one of the top 5 cyber threats around the world. IoT plays a vital role in today's connected world, spanning applications from smartwatches and IP cameras in traffic and road safety control, to CCTVs for smart city administration, PoS monitoring in retail stores, and inventory supply chain in manufacturing sector.

IoT architecture is based on three major components.

- **Edge client** – Collects data from connected devices and sends it to the gateway
- **Gateway** – Validates and forwards data to the cloud platform for storage and processing
- **Cloud** – Receives and processes data blocks from the gateway, stores them in cloud storage, and transfers for processing like analytics and reporting services hosted on the cloud native platform

Security design is vital in IoT architecture design strategy. We need to see this at four levels:

- **Infrastructure security** to manage security at the physical device level from device controllers, edge, gateway and cloud.
- **Platform security** to manage security at runtime services in edge and gateway in order to handle access control and protect data privacy requirements.
- **Network security** to manage, monitor and adopt data encryption during network transfers — two-way handshake to ensure secured clients are connected, and audit security logs for anomaly behaviour, if any.
- **Application security** to manage native and third party services handling data and to ensure that IoT services like API are protected from cyberattacks.

Potential challenges in IoT security could be data privacy compromised

when using third party services, unprotected data storage, lack of role-based access control (RBAC), weak cloud architecture to protect public exposed services, and insecure or outdated IoT device components being exposed to cyberattacks.

## Future developments in IoT enabled services

Here are some potential future developments in IoT-enabled services.

- **Ubiquitous connectivity:** 5G and beyond will enable massive scale IoT deployment. Connectivity will be faster, cheaper, and more reliable.
- **AI at the edge:** More advanced edge devices will allow localised AI inferencing, reducing latency and data transfer.
- **Increased adoption in industries:** Manufacturing, healthcare, smart cities and more will see expanded IoT adoption for monitoring, automation, etc.
- **Advanced analytics:** With more data, analytics and predictive modelling will become more sophisticated and accurate.
- **Growth of as-a-Service models:** Device management, connectivity, security, etc, will increasingly be offered 'as a service'.

- **New voice/gesture interfaces:** There will be more natural interfaces based on voice assistants, computer vision, and AR/VR.
  - **Increased focus on security:** Security architectures spanning edge, network, and cloud will be crucial.
  - **Distributed computing:** Swarms of collaborative edge devices for distributed computing paradigms will be leveraged.
  - **Personalised IoT:** Tailored solutions using wearables, smart home tech, and personalised healthcare will be popular.
  - **Sustainability:** Greener devices, renewable energy, and optimised energy usage will be in focus.
  - **Smart cities:** Urban IoT rollouts will improve traffic, public services, and more.
  - **Better hardware:** Smaller, cheaper, more efficient IoT sensors and edge hardware are on the cards.
  - **Closed loop automation:** Systems that translate IoT data into automated intelligent actions to improve business processes will be in common use.
- The future of IoT promises a brighter, more efficient, and interconnected world, but it will require overcoming scale, complexity, privacy and security challenges. **END** 🐧

## References

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